

CONSUMPTION, SAVINGS AND INVESTMENTS

Consumption Function

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$$C = \alpha + \beta Y$$

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However, the marginal increase in consumption will be less than the marginal increase in income.

That is $\beta < 1$.

The marginal increase in consumption resulting from an increase in income is called marginal propensity to consume (MPC).

MPC is the change in consumption arising from a unit change in income and is represented

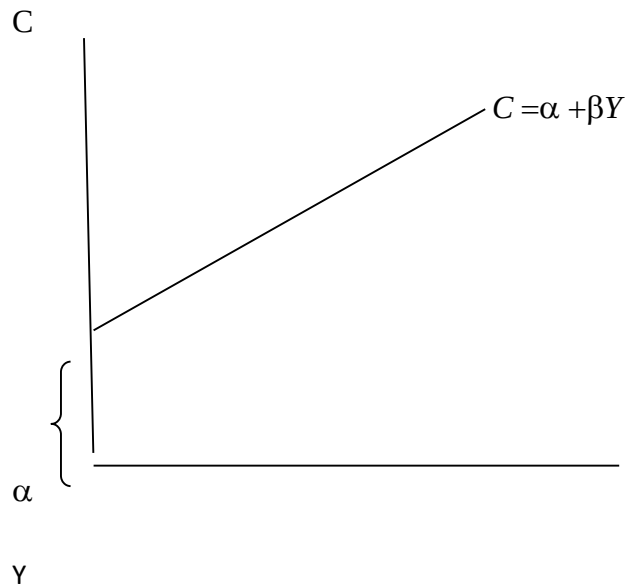
by

$$\beta = \Delta C / \Delta Y$$

The Consumption function is composed of autonomous consumption (α) and induced consumption (βY).

The autonomous part of consumption does not depend on disposable income. Thus it is the Consumption when income is zero. There is consumption at zero income because consumption also depends on other factors e.g. transfer payments and savings.

Graphically, the consumption function is presented as follows:



Average propensity to consume, (APC) is the proportion of disposable income that is spent on consumption. It is given by C/Y .

Therefore: from the above consumption function,

$$APC = \frac{C}{Y} = \frac{\alpha + \beta Y}{Y}$$

Determinants of Consumption

1. Income: The higher the income, the higher the consumption
2. Rate of Interest: According to classical economists, individuals will save more and spend less as interest rate increases
3. Relative Prices: This influences consumption behavior with consumers shifting to relatively cheaper goods.
4. Capital gains: According to Keynes, windfall gains/losses will influence consumption. Keynes argued that consumption of wealth owners can be influenced by sudden changes in the money value of their wealth. Sudden changes are common where the stock exchange market is composed of speculators.

5. Wealth: High stocks of wealth lead to low marginal value of wealth and hence less desire to accumulate more. As a result, this leads to increased consumption.
6. Money stock (Liquid assets): The higher the stock of liquid assets the higher the marginal propensity to consume.
7. Availability of consumer credit: Readily available and/or cheap consumer credit leads to consumers borrowing for consumption purposes. This pushes up the aggregate consumption function.

Theories of Consumption

Absolute Income Hypothesis

This hypothesis was postulated by Keynes.

According to this hypothesis, consumption is a function of current level of disposable personal income. Consumption is directly, but not proportionately related to current level of aggregate disposable income both in the short run and long run. This implies that C/Y decreases as income increases.

Keynes bases this assumption (disproportionate consumption change) on the argument that consumers' reaction to income change is not instant, but gradual since change in income may not be permanent.

Therefore, $C_t = \alpha + \beta Y_t + dC_{t-1}$.

This means that consumption over time (C_t) is not only dependent on income overtime (Y_t), but also previous level of consumption (dC_{t-1})

The Relative Income Hypothesis

This hypothesis was put forth by James Duesenberry in 1949.

It makes two assumptions:

- Consumption behavior of individuals is interdependent.
- Consumption relations are irreversible over time.

(i) Consumption behavior of individuals is interdependent.

This means that the ratio of income consumed depends on the individual's absolute income as well as their relative income, i.e. consumption will depend on the individual's percentile position in the total income distribution within a community.

(ii) Consumption relations are irreversible over time.

This assumption makes the relative income hypothesis to be referred to as previous peak theory. It means that when income falls during cyclical downswing the resultant fall in consumption will be less than proportionate because individuals base their consumption patterns on previous levels of income. When income increases, consumption increases proportionately, but when income falls below the previous peak, consumption does not fall proportionately.

Permanent Income Hypothesis

It states that consumption depends on permanent income. Permanent income is defined as the present value of the expected flow of long-term income.

According to this hypothesis, permanent consumption (C_P) is proportional to permanent income (Y_P), i.e. $C_P = f(Y_P)$.

The ratio of consumption (C_P) to induced consumption (βY_P) is constant at all levels of income. Income consists of two components namely: permanent income (Y_P) and transitory income (Y_T).

Life Cycle Income Hypothesis

It states that consumption is a function of the expected stream of disposable income over a long period of time and the present value of wealth.

Individuals are assumed to spread out the present value of all future income streams on consumption throughout their lifetime. Therefore, consumption is assumed to be a function of lifetime income.

SAVINGS FUNCTION

It describes the total amount of savings at each level of disposable personal income.

Savings is the difference between disposable income and consumption. Savings function is given by:

$$S = Y - C$$

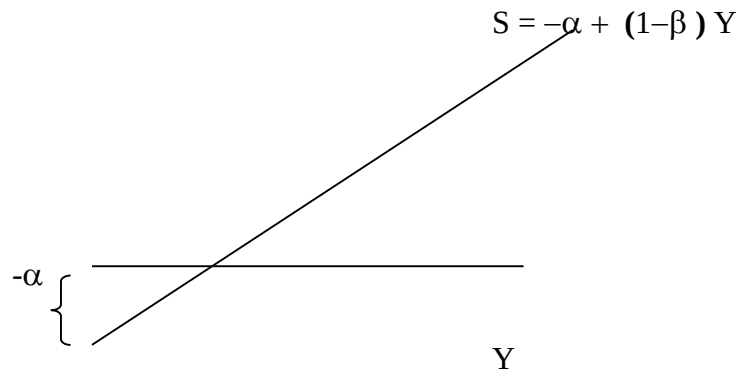
Given the consumption function, we can derive the savings function.

Suppose $C = \alpha + \beta Y$, then

$$S = Y - \alpha - \beta Y \Rightarrow S = -\alpha + Y - \beta Y,$$

Therefore, $S = -\alpha + (1 - \beta) Y$ is the savings function.

S



The savings function is upward sloping, implying that savings is an increasing function of income. The slope of the savings function is the marginal propensity to save (MPS).

MPS is the change in savings resulting from a unit change in personal disposable income.

The average propensity to save (APS) is the proportion of disposable personal income that is saved. It is given by S/Y , which implies that as income increases, APS decreases and vice versa.

Mathematically, the relationship between MPC and MPS is given as;

$$MPS = 1 - MPC \Rightarrow MPS + MPC = 1$$

INVESTMENT

Investment refers to the addition of capital stock in an economy. Therefore, it is given by the value of that part of aggregate output for any given year that takes the form of:

- Business fixed Investment which include plant and machinery, office buildings.
- Residential Construction Investment which includes houses and significant additions to Changes in firm's Inventories –as they provide revenue in the future, not today.
- Purchases of Durable Goods by Households for example cars, fridges.
- Government purchases of investment Goods and Building for example roads Hospital, harbors,
- Investment in Human Capital for example schooling university study, apprenticeships etc.
- Knowledge which. Includes things we know about scientific and other areas.

Types of Investment

a) Autonomous Investment (I_0):

This is investment that does not depend on the level of income. It is determined by exogenous functions e.g. inventories, population growth, wealth changes, research, etc.

B) Induced Investment:

This is investment that depends on income or profit. It is influenced by the factors, which affect income and profit e.g. prices, wages, interest, etc.

Induced investment is a function of income and is given by the equation

$$I = I_0 + \lambda Y,$$

Where λY = Induced Investment

λ = Marginal propensity to invest (MPI)

I_0 = Autonomous Investment

MPI is the change in investment due to a unit change in income i.e. $\Delta I/\Delta Y$, while

API is the ratio of investment to income i.e. I/Y .

c) Gross and Net Investment:

Gross investment is the total increase in capital stock in a year. Net investment is the net addition to capital stock in an economy after deducting capital consumption allowance from gross investment.

d) Intended and Unintended Investment

Intended (voluntary/planned) investment refers to deliberate accumulation of capital stock aimed at achieving a specific objective. Unintended (Involuntary/unplanned) investment is where capital stock accumulated due to unexpected fall in demand

e) Private Investment

It is investment made by private investors in an economy. It is normally made in response to profit expectations. It depends on the interest rate and the marginal efficiency of capital. It increases as the interest rate falls. It also increases as the marginal efficiency of capital (MEC) increases.

f) Public Investment

It is investment made by the government and other public enterprises.

g) Inventory investment

The last category of investment that we will study is inventory investment. It is not that large compared to the other forms of investment, but as mentioned at the beginning of this topic there is

a strong relationship between changes in inventory investment and changes in output over the business cycle.

Reasons for holding inventories

Before developing a theory of inventory investment we will study the reasons why inventories are held. Knowing the reasons for holding inventories will help us understand better the factors which influence the level of inventory investment. There are four reasons for holding inventories:

- 1) **Production smoothing:** often it is cheaper to produce goods at a steady rate, than to continually alter production runs. This means that during slow period's inventories build up and during boom periods inventories are run down.
- 2) **Inventories as a factor of production:** firms carry inventories of spare parts in case a machine breaks down, so that the entire assembly line does not come to a stop. E.g. car assembly factory.
- 3) **Stock out avoidance:** to avoid lost sales and profits when sales may be unexpectedly high e.g. a book retailer may carry multiple copies of a book if they are not sure how popular the book will be.
- 4) **Work in process:** partly completed productions are counted as inventories e.g. cheese making where cheeses are aging.

Determinants of Investment

1. Interest rate (i): Investment is inversely related to interest rate.
2. Internal rate of Return (IRR): It is the rate of interest that equates the present value of benefits from a project to the present value of its costs. A decision to invest is based on the comparison between IRR and i .
If $IRR > i$, investment is made
 $IRR < i$, no investment
 $IRR = i$, other factors are considered in deciding whether or not to invest.
3. Expected future income flows;- if the investor expects high profits, then investment will be undertaken and vice versa
4. Initial cost of the capital good and its useful life: - if the capital good is affordable then it will be purchased and vice versa. An investor will purchase a good that is likely to last longer
5. Degree of certainty: An investor considers the risks and uncertainties involved in a particular investment. if they are high he may not invest

6. Existing stock of capital: If the existing capital is large potential investors may be discouraged. Similarly if there is excess or idle capacity in existing capital stock, investment may be discouraged.

7. Level of income; a rise in the level of income in the economy due to rise in money wages and other factors prices raises the demand for goods and services and this in turn will induce an increase in investment.

8. Business expectations; if businessmen are optimistic and confident regarding future returns from capital goods they invest more.

9. Consumer demand; if the current demand for consumer goods is increasing rapidly, more investments will be made.